

In how many ways can we select a committee of two women and three men from a group of five distinct women and six distinct men?

Soln *major combination C*

The two women can be selected, $C(5, 2) = \frac{5 \times 4}{1 \times 2} = 10$ ways.

The three men can be selected, $C(6, 3) = \frac{6 \times 5 \times 4}{1 \times 2 \times 3} = 20$ ways.

By the multiplication Principle, the total number of committees = $10 \times 20 = 200$ ways.

From 21 consonants and 5 vowels, how many five-letter words can be formed with 3 consonants and 2 vowels if (all the letters of each word are different)?

Soln

3 consonants can be chosen from 21 consonants in

$$C(21, 3) = \frac{21 \times 20 \times 19}{1 \times 2 \times 3} = 1330 \text{ ways}$$

2 vowels can be chosen from

$$5 \text{ vowels in } C(5, 2) = \frac{5 \times 4}{1 \times 2} = 10 \text{ ways}$$

The 5 letters in each word

may be *order / Arrange* permuted among themselves in $5P_5 = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ ways

The total number of words = $1330 \times 10 \times 120 = 1596000$ ways #

Determine how many strings can be formed by ordering the letters ABCDE subject to the conditions given.

i) Contains the *not before* substring ACE.

Soln

$$\boxed{ACE} \quad \boxed{B} \quad \boxed{D}$$

The number of ways contain the substring ACE = ${}_3P_3 = 3! = 3 \times 2 \times 1 = 6$ ways

ii) Contains the letters ACE together in any order.

Soln

$$\boxed{A} \quad \boxed{C} \quad \boxed{E} = 3! = 3 \times 2 \times 1 = 6$$

$$\boxed{ACE} \quad \boxed{B} \quad \boxed{D} = 3! = 6$$

The number of ways contain the letters ACE together in any order = $3! \times 3! = 3 \times 2 \times 1 \times 3 \times 2 \times 1 = 36$ ways

iii) Contain either the substring AE or the substring EA.

$$\boxed{AE} \quad \boxed{B} \quad \boxed{C} \quad \boxed{D}$$

The number of ways contains the substring AE = $4!$

The number of ways contains the substring EA = $4!$

∴ The number of ways contain either the substring AE or the substring EA = $4! + 4!$

$$= 4 \times 3 \times 2 \times 1 + 4 \times 3 \times 2 \times 1$$

$$= 24 + 24$$

$$= 48 \text{ \# ways}$$

Contains the substring DB and AE.

Soln

DB AE C

The number of ways contain the substring DB and AE = $3! = 3 \cdot 2 \cdot 1 = 6$ ways

Contains neither of the substring AB, CD.

Soln

Total number of ways = $5! = 120$ ways

The number of ways for substring AB = $4! = 24$

The number of ways for substring CD = $4! = 24$

The number of ways contain AB and CD = $3! = 6$

The number of ways for AB or CD

= number of ways for substring AB + number of ways for substring CD - number of ways contain AB and CD

$$= 24 + 24 - 6 = 42 \text{ ways}$$

The number of ways neither of AB, CD

= total number of ways

- number of ways AB or CD

$$= 120 - 42$$

$$= 78 \text{ \# ways}$$

Let $\alpha = 132$ and $\beta = 1324$ be strings over $\{1, 2, 3, 4\}$.

Soln

By definition, $p = 3, q = 4$

$$s_1 = 1, s_2 = 3, s_3 = 2$$

$$t_1 = 1, t_2 = 3, t_3 = 2, t_4 = 4$$

$$p = 3, q = 4$$

$$p < q \text{ and}$$

$$s_i = t_i, i = 1, 2, 3$$

Condition (a) satisfied.

$$\alpha < \beta$$

Let $\alpha = 13246$ and $\beta = 1342$ be strings over $\{1, 2, 3, 4, 5, 6\}$.

Soln

By definition, $p = 5, q = 4$

$$s_1 = 1, s_2 = 3, s_3 = 2, s_4 = 4, s_5 = 6$$

$$t_1 = 1, t_2 = 3, t_3 = 4, t_4 = 2$$

The smallest i for which

$$s_i \neq t_i \text{ is } i = 3$$

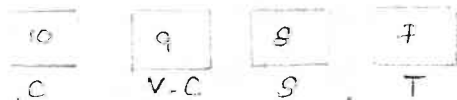
$$s_3 < t_3$$

Condition (b) satisfied.

$$\text{Hence } \alpha < \beta. \text{ \#}$$

In how many ways can we select a chairperson, vice-chairperson, secretary, and treasurer from a group of 10 persons?

Soln



The number of ways select a chairperson = 10 ways

The number of ways select vice-chairperson = 9 ways

The number of ways select secretary = 8 ways

The number of ways select treasurer = 7 ways

∴ The number of permutations of four element select from a group of 10 persons

$$= P(10, 4) = \frac{10!}{(10-4)!}$$

$$= \frac{10!}{6!} = \frac{10 \times 9 \times 8 \times 7 \times 6!}{6!}$$

$$= 5040 \text{ ways}$$

In how many ways can we seven men and five women wait in line if no two women stand together?

Soln

We will consider line up the men and women by a two-step process. Line up the men, line up the women.

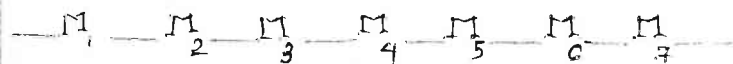
The men can line up in

$$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$$

Once the men have lined up, e.g., in positions M_1 up to M_7 ,

the women have eight possible positions in which to stand, indicated by blanks since no two women

stand together.



Thus the women can stand in $P(8, 5) = 8 \times 7 \times 6 \times 5 \times 4 = 6720$ ways

By the Multiplication Principle, the number of ways seven distinct men and five distinct women can wait in line if no two women stand together is

$$5040 \times 6720 = 33,868,800 \text{ ways}$$

How many arrangements of the letters of the word COINS can be made if every arrangement is to begin and end with the vowels?

Soln

There are 2 vowels and 3 consonants in the word COINS.

The 3 consonants C, N, S can be arrangement among themselves in ${}_3P_3 = 3! = 3 \times 2 \times 1 = 6$ ways. Since an arrangement may begin with either vowel and end with the other, the number of vowels in which the vowels may be arranged is 2.

∴ The required number of arrangements = $6 \times 2 = 12$ ways

Find the permutation that will be generated by Algorithm after the permutation 1 2 3 5 4.

Soln

1 2 3 5 4

$$s_1 = 1, s_2 = 2, s_3 = 3, s_4 = 5, s_5 = 4$$

$$s_1 < s_2$$

$$s_2 < s_3$$

$$s_3 < s_4$$

The largest index m satisfying

$$s_m < s_{m+1} \text{ is } 3, m = 3.$$

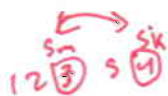
$$s_4 > s_3$$

$$s_5 > s_3$$

The largest index k satisfying

$$s_k > s_m \text{ is } k = 5.$$

Swap s_m and s_k .



$$s = 1 2 4 5 3$$

Reverse the order $s_4 s_5 = 35$

We obtain the desired permutation
on 1 2 4 35.

Find the permutation that will be generated by Algorithm after the permutation 625431.

Soln

6 2 5 4 3 1

$$s_1 = 6, s_2 = 2, s_3 = 5, s_4 = 4,$$

$$s_5 = 3, s_6 = 1.$$

$$s_2 < s_3$$

The largest index m satisfying

$$s_m < s_{m+1} \text{ is } 2, m = 2.$$

$$s_1 > s_2$$

$$s_3 > s_2$$

$$s_4 > s_2$$

$$s_5 > s_2$$

The largest index k satisfying

$$s_k > s_m \text{ is } k = 5.$$

Swap s_m and s_k .

$$s = 6 3 5 4 2 1$$

Reverse the order $s_3 s_4 s_5 s_6$
 $= 1245$

We obtain the desired permutation 631245.

Find the permutation that will be generated by Algorithm after the permutation 12876543.

Soln

1 2 8 7 6 5 4 3

$$s_1 = 1, s_2 = 2, s_3 = 8, s_4 = 7, s_5 = 6,$$

$$s_6 = 5, s_7 = 4, s_8 = 3.$$

$$s_1 < s_2$$

$$s_2 < s_3$$

The largest index m satisfying

$$s_m < s_{m+1} \text{ is } 2, m = 2.$$

$$s_3 > s_2$$

$$s_4 > s_2$$

$$s_5 > s_2$$

$$s_6 > s_2$$

$$s_7 > s_2$$

$$s_8 > s_2$$

The largest index k satisfying

$$s_k > s_m \text{ is } k = 8.$$

Swap s_m and s_k

$$s = 1 3 8 7 6 5 4 2$$

Reverse the order

$$s_3 s_4 s_5 s_6 s_7 s_8 = 245678$$

We obtain the desired permutation 13245678.

mult. given number by 2 to get the final result

inventory consists of a list of 115 items, each marked "available" or "unavailable".

There are 60 available items. Show that there are at least two available items in the list exactly four items apart.

Proof

Let a_i be the i th available item. Consider the numbers $a_1, a_2, \dots, a_{60}$ and $a_1 + 4, a_2 + 4, \dots, a_{60} + 4$. — (2)

The 120 numbers in eq (1) and eq (2) have possible values only from 1 to 119.

By the Pigeonhole Principle (2nd form) at least two of these values coincide.

The numbers of eq (1) are all distinct and hence the numbers of eq (2) are also all distinct.

The same number in (1) is equal to some number in (2). $a_i = a_j + 4$ for some i and j .

Ten persons have first names Age, Bo and Chit and last names Hla, Maung and San. Show that at least two persons have the same first and last names.

Soln

First names

Age

Bo

Chit

Last Names

Hla

Maung

San

By multiplication principle, The number of possible names = $3 \times 3 = 9$

$n = \text{persons} = 10$, $k = \text{names} = 9$ (5) (pigeons) (pigeonhole)

By the Pigeonhole principle (1st form), at least one pigeonhole contains at least two pigeons.

Thus, there must be at least two persons have the same full names.

Show that every set of 11 shoes chosen from among 10 pairs of shoes contain at least one matched pair.

Soln

Shoes (Pigeons) Pairs (Pigeonhole)

Assign each of the 11 selected shoes to the hole (pair) it belongs to.

Since there are 11 shoes but only 10 possible pairs, the Pigeonhole Principle states at least one hole must contain more than one pigeon.

Because each hole (pair) can only hold a maximum of two shoes, a left and a right, having 'more than one' shoe in a hole means that hole contains exactly two shoes.

Two shoes from the same pair constitute a matched pair.

∴ Any set of 11 shoes from 10 pairs must contain at least one matched pair.

How many arrangements of the letters of the 'WOMEN' can be made if every arrangement is to begin and end with the vowels?

Soln

The word 'WOMEN'.

For the 1st letter must be O and E.

The number of arrangement begin with vowels = 2 ways.

For the last letter to be

vowel must be O and E
The number of arrangement end with vowels = 2 ways

The number of arrangement = $3! = 6$ ways

∴ The number of arrangement begin with vowel and ends with vowel

$$= 2 \times 2 \times 6$$

$$= 24 \text{ ways} \#$$

Solve the recurrence relation

$a_n = 7a_{n-1} - 10a_{n-2}$ for the initial conditions $a_0 = 5, a_1 = 16$.

Soln

$$a_n = 7a_{n-1} - 10a_{n-2}, a_0 = 5$$

$$a_1 = 16$$

Let $a_n = t^n$

$$t^n = 7t^{n-1} - 10t^{n-2}$$

Dividing t^{n-2} ,

$$t^2 = 7t - 10$$

$$t^2 - 7t + 10 = 0$$

$$(t-2)(t-5) = 0$$

$$t-2=0 \quad (\text{or}) \quad t-5=0$$

$$t=2 \quad (\text{or}) \quad t=5$$

$$a_n = b r_1^n + d r_2^n \quad \leftarrow \text{general form}$$

$$= b \cdot 2^n + d \cdot 5^n$$

If $a_0 = 5 \Rightarrow a_0 = b \cdot 2^0 + d \cdot 5^0$

$$5 = b + d \quad \text{--- (*)}$$

If $a_1 = 16 \Rightarrow a_1 = b \cdot (2)^1 + d \cdot (5)^1$

$$16 = 2b + 5d$$

$$\text{Eq (*)} \times 2 \Rightarrow 10 = 2b + 2d$$

$$6 = 3d$$

$$d = 2$$

Substituting in eq (*),

$$b + d = 5$$

$$b = 3$$

$$\therefore a_n = 3 \times 2^n + 2 \times 5^n \#$$

The three consonants are W, M and N.

person invests initial 2000K at 14 percent compounded annually. Let A_n represent the amount at the end of n years.

- (i) Find a recurrence relation for the sequence $A_0, A_1, \dots$
- (ii) Find an initial condition for the sequence $A_0, A_1, \dots$
- (iii) Find A_1, A_2 and A_3 .
- (iv) Find an explicit formula for A_n .
- (v) How long will it take for a person to double the initial investment?

Soln

$A_0 = 2000K$

$r = 14\% = 0.14$

Let A_n be the amount at the end of n years. Formula

i) $A_n = A_{n-1} + A_{n-1} (0.14)$
 $= 1.14 A_{n-1}, n \geq 1$

ii) The initial condition $= A_0 = 2000K$

iii) $A_n = A_{n-1} + A_{n-1} (0.14), n \geq 1$

$A_1 = A_0 + A_0 (0.14) = 1.14 A_0$
 $= 2280$

$A_2 = A_1 + A_1 (0.14) = 1.14 A_1 = 2599.2$

$A_3 = A_2 + A_2 (0.14) = 1.14 A_2 = 2963.088$

iv) $A_n = ?$

$A_1 = 1.14 A_0$

$A_2 = 1.14 A_1$

$= (1.14) (1.14) A_0$

$= (1.14)^2 A_0$

$A_3 = 1.14 A_2$

$= (1.14) (1.14)^2 A_0$

$= (1.14)^3 A_0$

$A_n = (1.14)^n A_0 = (1.14)^n (2000)$

v) $A_n = 2 A_0$

$(1.14)^n A_0 = 2 A_0$

$(1.14)^n = 2$

$\log (1.14)^n = \log 2$

$n \log (1.14) = \log 2$

$n = \frac{\log 2}{\log 1.14}$

$n = \frac{0.3010}{0.0569} = 5.29 \text{ years}$

A person invests 2000K each year in a tax-sheltered annuity at 10 percent compounded annually. Let A_n represent the amount at the end of n years.

i) Find a recurrence relation for the sequence $A_0, A_1, \dots$

ii) Find an initial condition for the sequence $A_0, A_1, \dots$

iii) Find A_1, A_2 and A_3 .

iv) Find an explicit formula for A_n .

Soln

$A_0 = 2000 K$

$r = 10\% = 0.1$

Let A_n be the amount at the end of n years.

i) $A_n = A_{n-1} + A_{n-1} (0.1)$
 $= 1.1 A_{n-1}, n \geq 1$

ii) The initial condition $= A_0 = 2000 K$

Recurrence relation
 Initial condition
 Find terms
 Find explicit formula

$$\text{iii)} \quad A_0 = 2000$$

$$A_n = A_{n-1} + A_{n-1} (0.1), \quad n \geq 1$$

$$A_1 = A_0 + A_0 (0.1)$$

$$= 1.1 A_0 = 2200$$

$$A_2 = A_1 + A_1 (0.1) = 1.1 A_1$$

$$= 2420$$

$$A_3 = A_2 + A_2 (0.1) = 1.1 A_2$$

$$= 2662$$

$$\text{iv)} \quad A_n = A_{n-1} + A_{n-1} (0.1) = 1.1 A_{n-1}$$

$$A_1 = (1.1) A_0$$

$$A_2 = (1.1) A_1 = (1.1)^2 A_0$$

$$A_3 = (1.1) A_2 = (1.1)^3 A_0$$

$$A_n = (1.1) A_{n-1} = (1.1)^n A_0$$

$$= (1.1)^n (2000)$$

A person invests 3000 K at 12 percent annual interest compounded quarterly. Let A_n represents the amount at the end of n years.

i) Find a recurrence relation for the sequence $A_0, A_1, \dots$

ii) Find an initial condition for the sequence $A_0, A_1, \dots$

iii) Find A_1, A_2 and A_3 .

iv) Find an explicit formula for A_n .

v) How long will it take for a person to double the initial investment?

Soln

$$A_0 = 3000 \text{ K}$$

$$r = \frac{12\%}{4} = 3\% = 0.03$$

Let A_n be the amount at the end of n years.

$$\text{i)} \quad A_n = A_{n-1} + A_{n-1} (0.03)$$

$$= (1.03) A_{n-1}, \quad n \geq 1$$

$$\text{ii)} \quad \text{The initial condition} = A_0 = 3000 \text{ K}$$

$$\text{iii)} \quad A_0 = 3000$$

$$A_n = A_{n-1} + A_{n-1} (0.03), \quad n \geq 1$$

After 3 months, total amount

$$= A_0 + A_0 (0.03) = 1.03 A_0$$

After 6 months, total amount

$$= (1.03)^2 A_0$$

After 9 months, total amount

$$= (1.03)^3 A_0$$

After 12 months, total amount

$$\text{at the end of 1 yr} = A_1 = (1.03)^4 A_0$$

Total amount at the end of 2 yr,

$$A_2 = (1.03)^8 A_0$$

Total amount at the end of 3 yr,

$$A_3 = (1.03)^{12} A_0$$

iv) Total amount at the end of

$$n \text{ yr}, \quad A_n = (1.03)^{4n} A_0$$

$$\therefore A_1 = (1.03)^{4 \times 1} A_0$$

$$A_2 = (1.03)^{4 \times 2} A_0$$

$$A_3 = (1.03)^{4 \times 3} A_0$$

$$A_0 = 3000$$

$$A_n = 2 A_0$$

$$(1.03)^{4n} A_0 = 2 A_0$$

$$(1.03)^{4n} = 2$$

$$\log (1.03)^{4n} = \log 2$$

$$4n \log 1.03 = \log 2$$

$$4n = \frac{\log 2}{\log 1.03}$$

$$4n = \frac{0.3010}{0.0128} = 23.516$$

$$\therefore n = 5.88 \text{ years}$$